

[1]

✓ Os

```
import numpy as np

products = {
    101: [1, 1, 0], 102: [1, 1, 0], 103: [0, 0, 1],
    104: [1, 1, 1], 105: [0, 0, 1], 106: [1, 1, 0], 107: [0, 0, 1]
}
purchased_ids = [103]
clicked_ids = [104, 105, 106]

affinity = np.array([0, 0, 0])
for pid in purchased_ids:
    affinity += np.array(products[pid]) * 3
for pid in clicked_ids:
    affinity += np.array(products[pid])

print("Affinity Profile =", affinity)
```

▼

```
... Affinity Profile = [2 2 5]
```

[2]

✓ Os

```
scores = {}
for pid, feature in products.items():
    if pid not in purchased_ids:
        scores[pid] = np.dot(feature, affinity)

top = sorted(scores.items(), key=lambda x: x[1], reverse=True)

print("Top 3 Recommendations:")
for p in top[:3]:
    print(p)
```

▼

```
... Top 3 Recommendations:
(104, np.int64(9))
(105, np.int64(5))
(107, np.int64(5))
```

[3]
✓ 0s

```
memory = []  
sequence = [106, 104, 105]  
  
for pid in sequence:  
    memory.append(pid)  
    if len(memory) > 2:  
        memory.pop(0)  
  
    score = np.array([0, 0, 0])  
    for m in memory:  
        score += np.array(products[m])  
  
    categories = ["Electronics", "Gadgets", "Fitness"]  
    print("Memory:", memory)  
    print("Current Interest:", categories[np.argmax(score)])
```

```
... Memory: [106]  
Current Interest: Electronics  
Memory: [106, 104]  
Current Interest: Electronics  
Memory: [104, 105]  
Current Interest: Fitness
```